

# Password Manager System

## Detailed Assignment Report with Explanation

### Introduction

The Password Manager System is a Java-based desktop application developed using Object Oriented Programming concepts. In modern digital life, users create multiple online accounts for websites, social media, banking, and educational portals. Remembering different usernames and passwords becomes difficult, and many users store passwords insecurely or reuse weak passwords.

This project solves the problem by providing a centralized system where users can securely store and manage their credentials. The application uses Java Swing for graphical interface design, JDBC for database connectivity, and MySQL as the backend database. Passwords are encoded using Base64 encoding before storing them in the database.

### Objectives of the Project

The major objectives of the project are:

- To create a secure desktop application for password management.
- To implement user registration and login functionality.
- To securely store website credentials in a database.
- To implement JDBC connectivity with MySQL.
- To learn practical implementation of Java OOP concepts.
- To create a user-friendly GUI using Java Swing.

This project also helps students understand real-world software development practices.

### Technologies Used

**Java:** Used as the main programming language for developing the application.

**Java Swing:** Used to create graphical user interfaces like login forms, registration windows, and dashboard screens.

**JDBC:** JDBC (Java Database Connectivity) is used to connect Java applications with MySQL database.

**MySQL:** Used to store user details and credentials permanently.

**Base64 Encoding:** Used for basic password encryption before storing passwords.

### System Architecture

The project follows a modular architecture where different tasks are divided into separate packages and classes.

**DAO Package:** Handles all database operations such as insert, update, and retrieval.

**Model Package:** Contains classes representing entities such as User and Credential.

**UI Package:** Contains all GUI screens created using Swing.

**Utility Package:** Contains helper classes such as DBConnection and Encryption.

**Main Package:** Contains the main entry point of the application.

## Explanation of Important Classes

### **MainApp.java**

This is the starting point of the application. It launches the login screen when the program starts.

### **LoginFrame.java**

This class creates the login window where users enter username and password. It validates user credentials by checking records from the database.

### **RegisterFrame.java**

This class allows new users to register into the system. It stores new user details into the MySQL database.

### **DashboardFrame.java**

This is the main dashboard shown after successful login. Users can add website credentials and view stored credentials here.

### **UserDAO.java**

This class performs database operations related to users such as registration and login verification.

### **CredentialDAO.java**

This class manages insertion and retrieval of credentials from the database.

### **DBConnection.java**

This class creates and manages the connection between Java application and MySQL database.

### **Encryption.java**

This class handles password encoding and decoding using Base64.

## Database Design

The application uses a MySQL database named **password\_manager**.

### **Users Table:**

Stores user account details.

Fields:

- id
- username
- password

### **Credentials Table:**

Stores website credentials.

Fields:

- id
- user\_id
- website
- site\_username
- site\_password

The database provides permanent storage for application data.

## JDBC Connectivity

JDBC is used to establish communication between Java application and MySQL database.

Steps involved in JDBC connectivity:

1. Load JDBC Driver
2. Establish Connection
3. Create PreparedStatement
4. Execute SQL Query
5. Process ResultSet
6. Close Connection

PreparedStatement is used because it improves security and prevents SQL Injection attacks.

## Encryption Mechanism

The application uses Base64 encoding for basic password security.

Before storing passwords in the database:

- Passwords are converted into encoded format.
- Encoded data is stored in database.

When passwords are displayed:

- Encoded passwords are decoded back into original form.

This prevents direct visibility of passwords in database records.

## Features of the Application

- User Registration
- User Login Authentication
- Website Credential Storage
- Password Encryption
- Graphical User Interface
- MySQL Database Integration
- Easy-to-use Dashboard
- Secure Credential Management

## Advantages of the System

The Password Manager System provides multiple benefits:

- Helps users manage multiple passwords easily.
- Reduces chances of forgetting passwords.
- Centralized storage for credentials.
- Beginner-friendly graphical interface.
- Practical implementation of Java concepts.
- Improves understanding of database connectivity and software development.

## Limitations of the System

Although the system works successfully, it has some limitations:

- Uses only basic Base64 encoding instead of advanced encryption.
- No cloud synchronization support.
- No password recovery option.
- No multi-factor authentication.
- Desktop-only application.

## Future Enhancements

The project can be improved further using advanced features such as:

- AES Encryption
- OTP Verification
- Password Strength Checker
- Cloud Storage
- Mobile Application Version
- Auto Password Generator
- Biometric Authentication
- Backup and Restore Functionality

## Testing

Different types of testing were performed during development:

**Unit Testing:** Individual classes and methods were tested separately.

**Integration Testing:** Interaction between GUI and database was tested.

**Functional Testing:** Login, registration, and credential management were verified.

**Exception Testing:** Invalid inputs and database errors were handled properly.

## Conclusion

The Password Manager System successfully demonstrates implementation of a Java desktop application using Object Oriented Programming concepts. The project integrates Java Swing, JDBC, MySQL, and password encryption techniques to provide secure credential management.

Through this project, students gain practical knowledge of GUI design, database operations, event handling, exception handling, and software architecture. The system can further be enhanced into a professional-level password management application with advanced security features.